

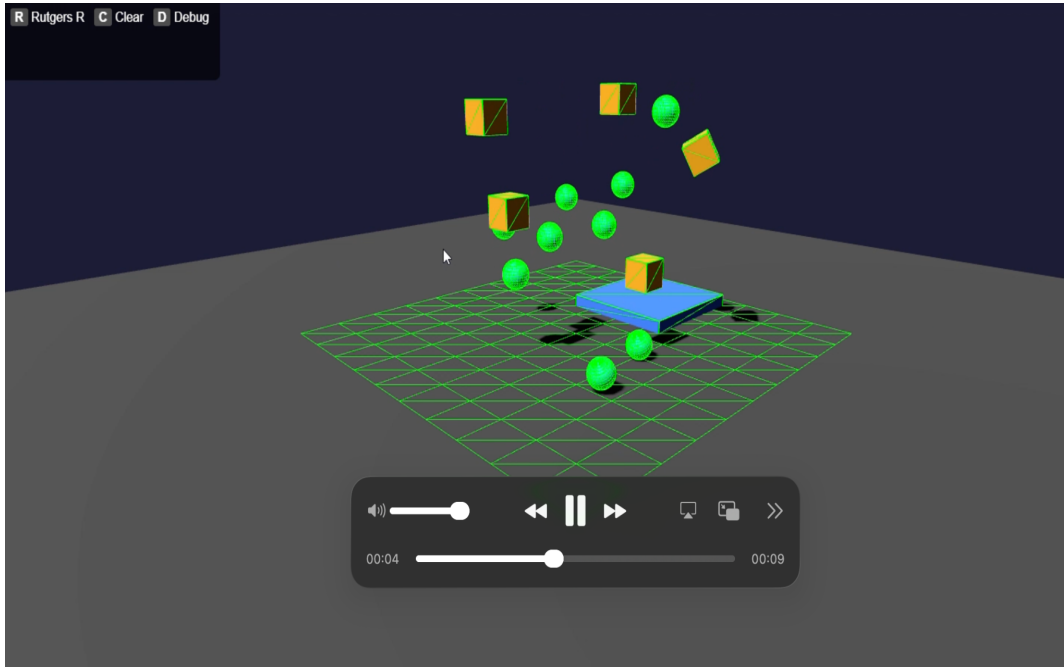
Homework 4: Physics Simulation Report

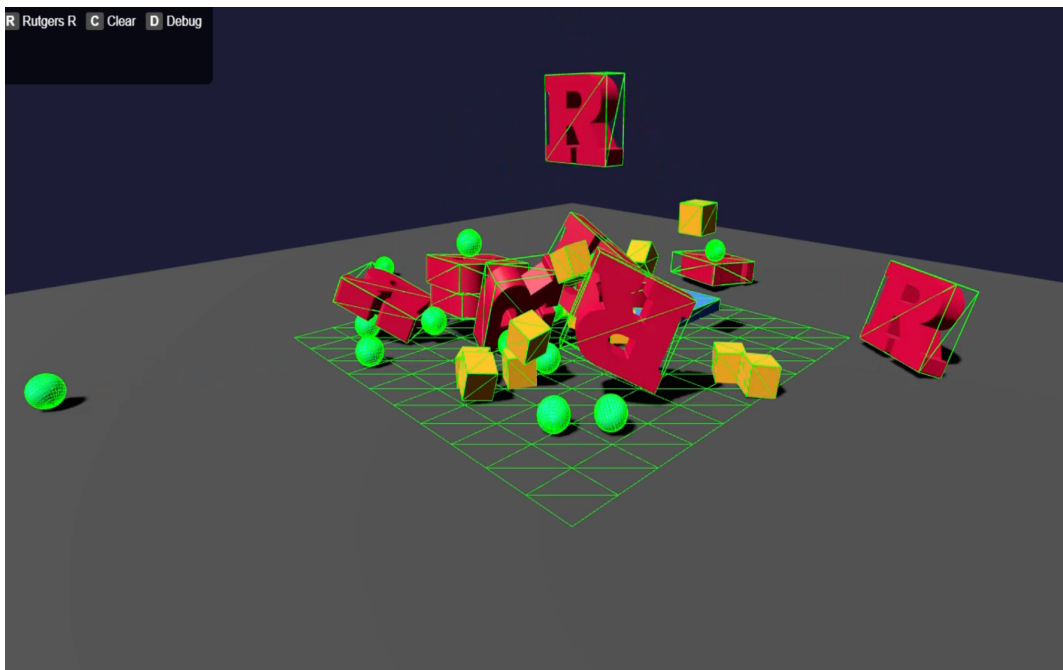
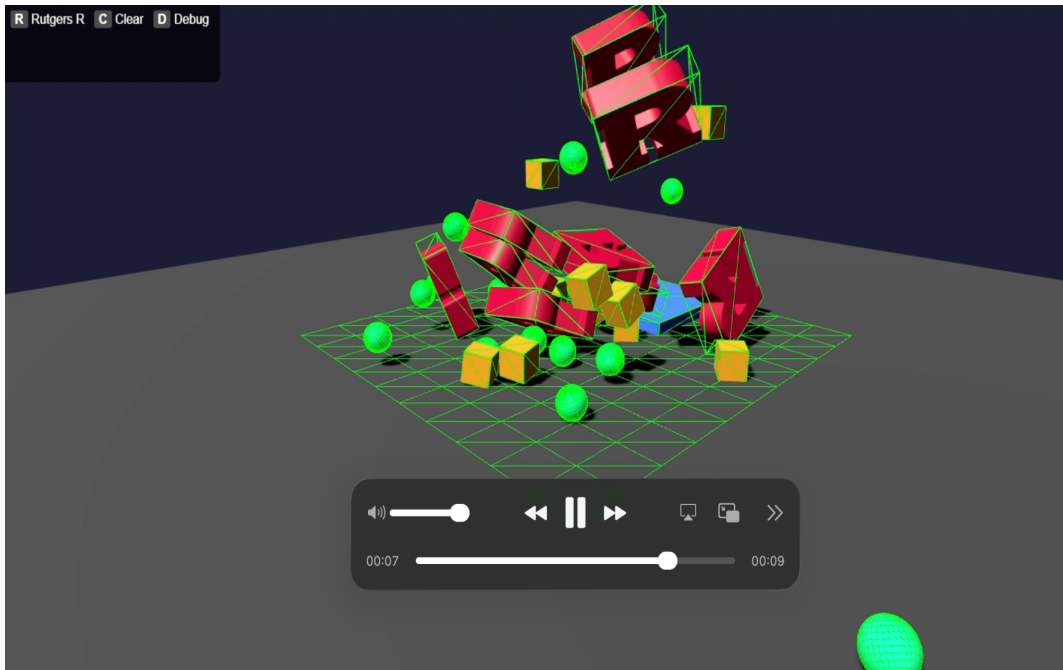
Task 1: Bug Fixes

- Fixed box physics mismatch by correcting half-extents.
- Fixed sphere physics using correct radius.
- Adjusted platform position to align with the floor.
- Ensured mesh and physics bodies stay synchronized using position and quaternion updates.

Task 2: Physics Scene

- Implemented Rutgers R spawning using GLTFLoader.
- Used bounding box to calculate physics body dimensions.
- Verified realistic collisions between R, spheres, and cubes.





Task 3: VR Interaction

- Implemented object selection using trigger input.
- Implemented object spawning using grip/squeeze input.
- Ensured spawned objects are added to the interaction group and can be dragged.

